

Department of Electrical & Electronic Engineering

Brac University

Semester:Summer 25



Inspiring Excellence

EEE203L

Electronic Circuits II Laboratory

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Section:02

Experiment No. 5

Familiarization with Passive Filters

Objective:

The purpose of this experiment is to study passive filters (specially low pass and high pass filters) and understand how they allow certain frequencies to pass while blocking others. By constructing RC filter circuits, measuring their output at different frequencies and comparing the results with theoretical cutoff values, we will verify the frequency selective behavior of passive filters.

Theoretical Background:

A filter is a circuit that passes some frequency component of a signal and alternates others.

In passive filters only resistores(R), capacitores (c) and inductores (L) are used.

Two main types of experiment are visible.

- Low pass filters(LPF)
- High pass filter(HPF)

For the RC circuit, the cutoff frequency is

$$f_c = 1/2\pi RC$$

At f_c , the output voltage drops to 0.707 of its maximum value(or the power drops to half).

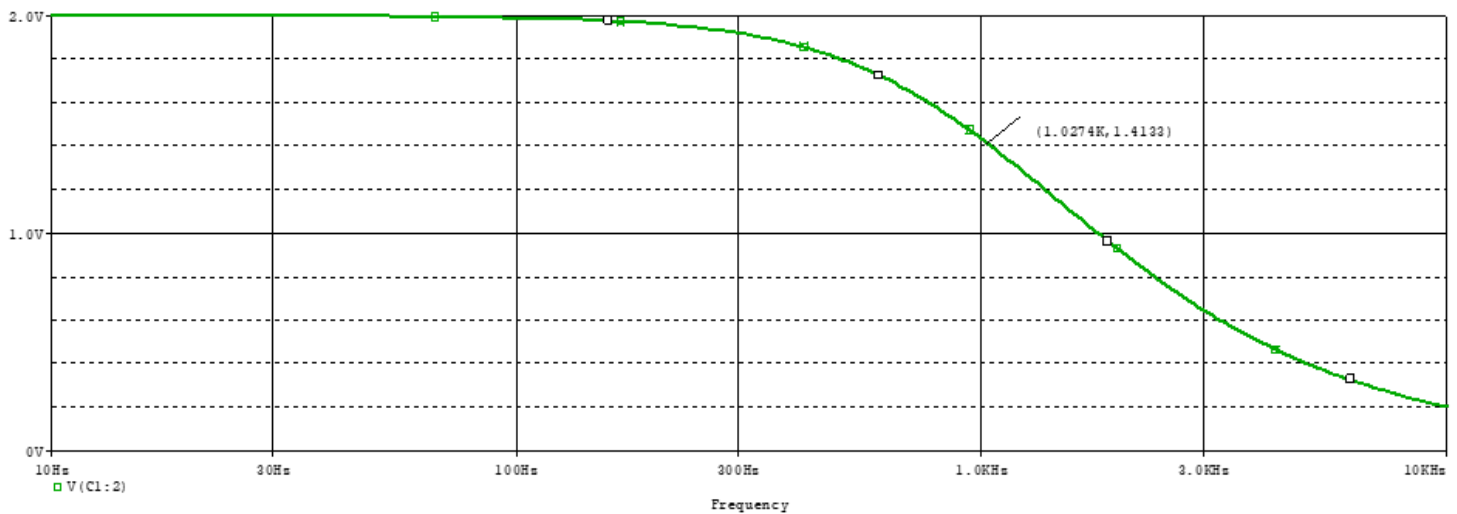
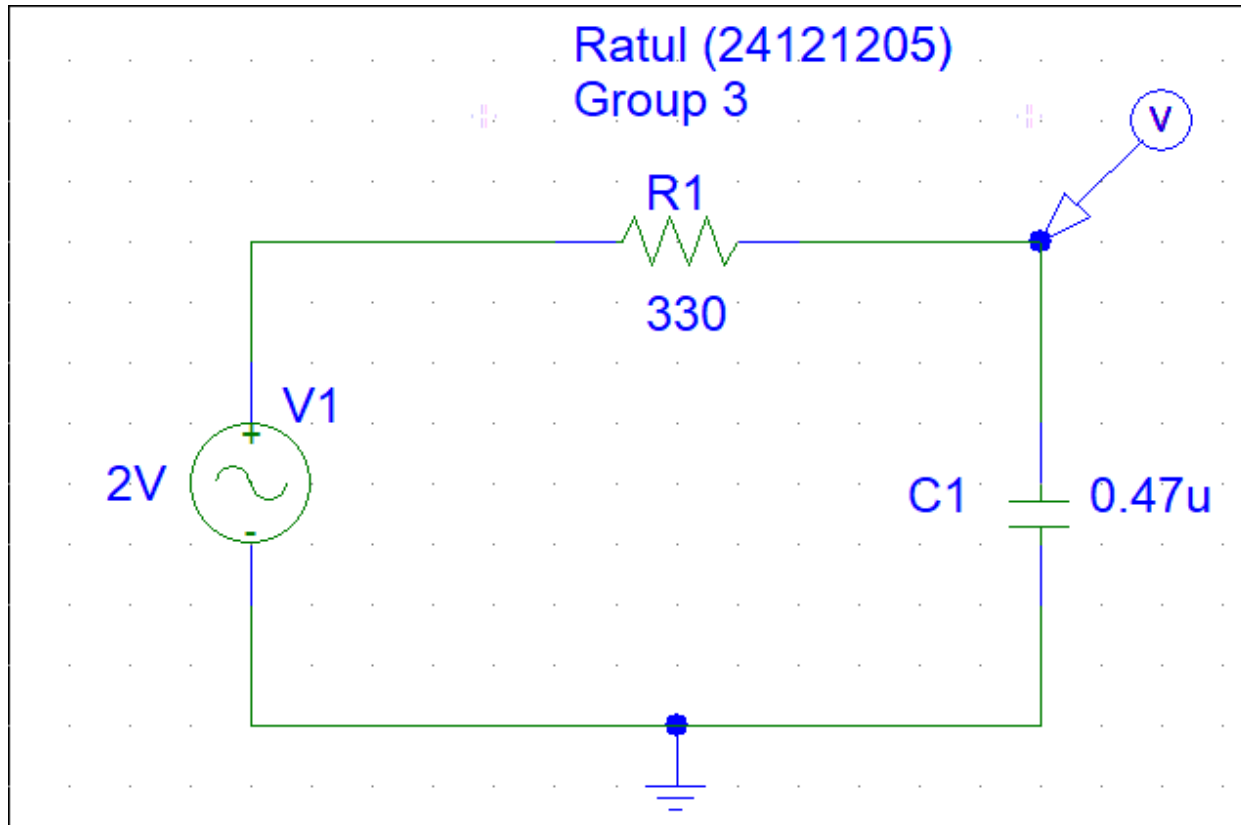
In AC analysis

- At low frequencies a capacitor offers high reactance $x_c = 1/2\pi fC$
- At high frequencies x_c becomes small allowing more current to pass.

By swaping the input frequency and measuring the output, we can observe the filters gain vs frequency behavior and verify f_c experimentally.

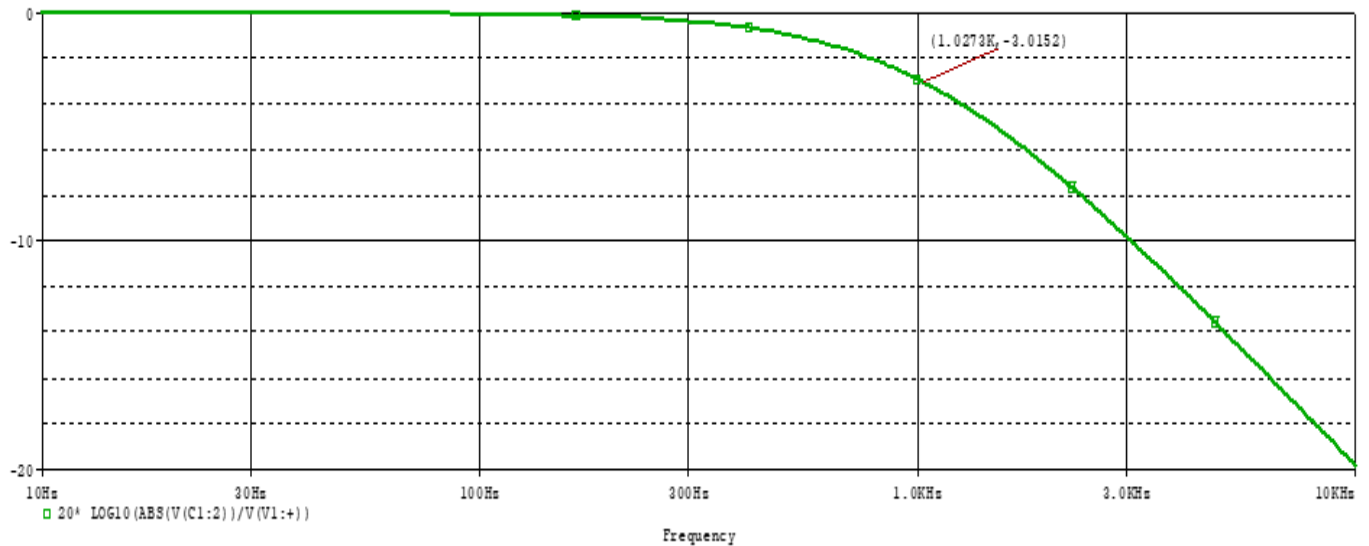
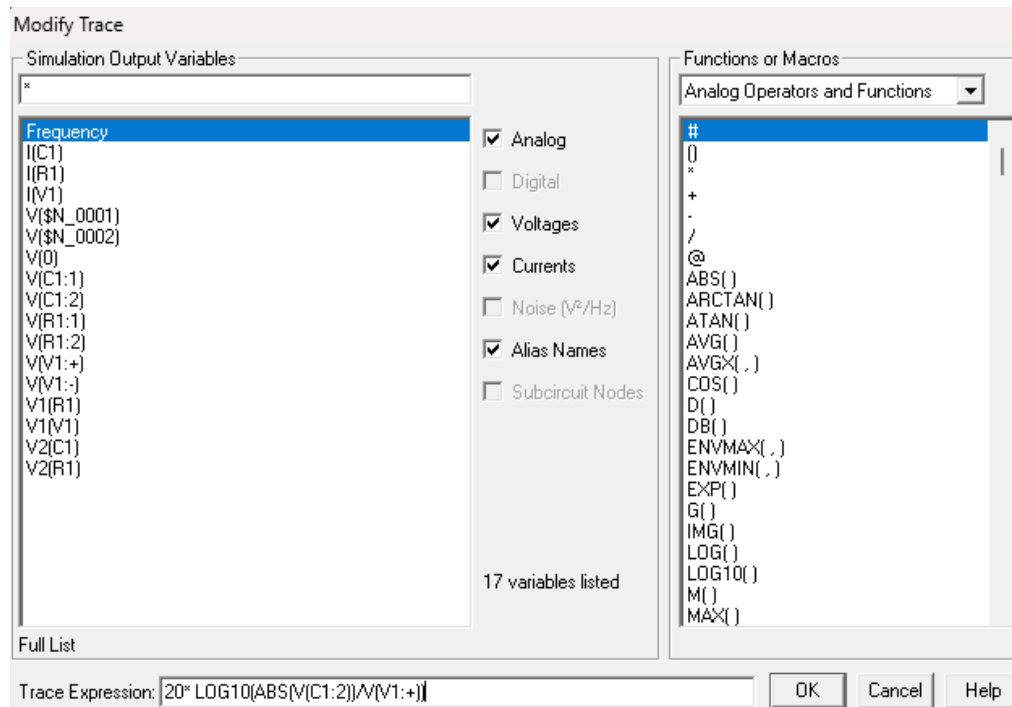
Circuit Diagram:

Low pass Filter

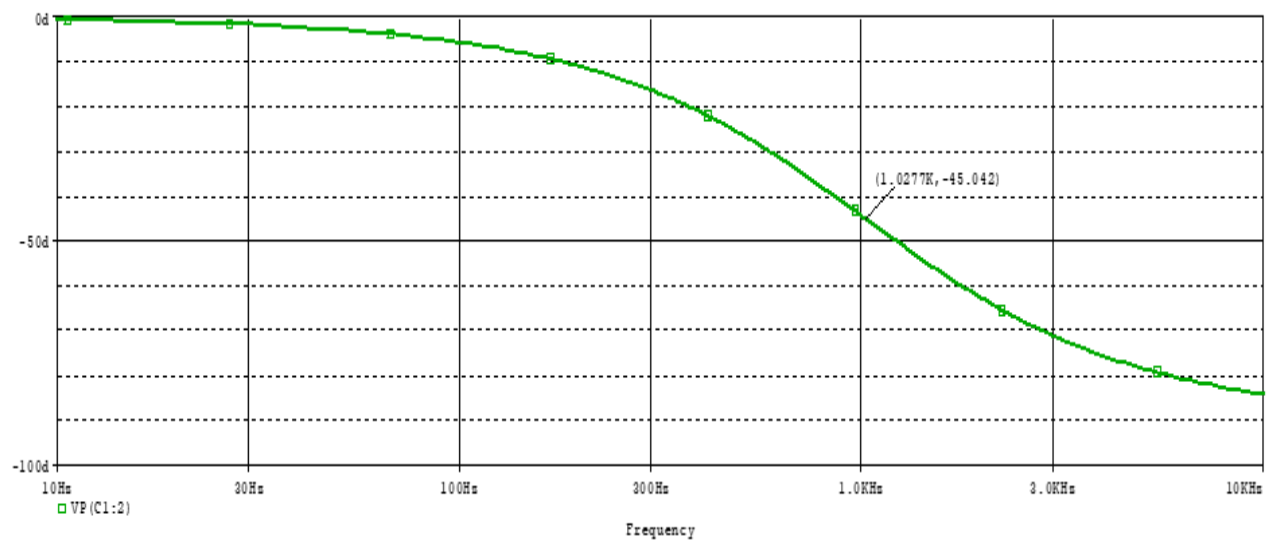
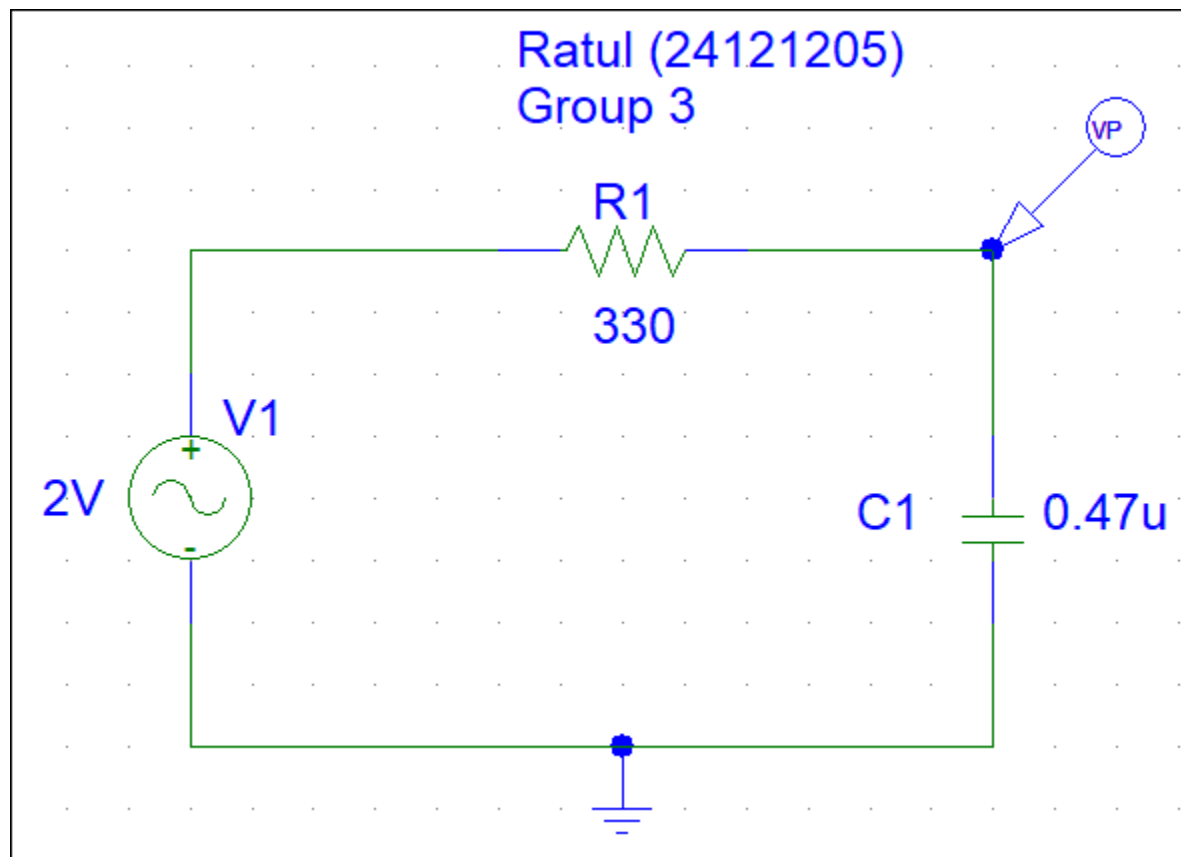


In here, Cutoff = $2 \times 0.707 = 1.414$

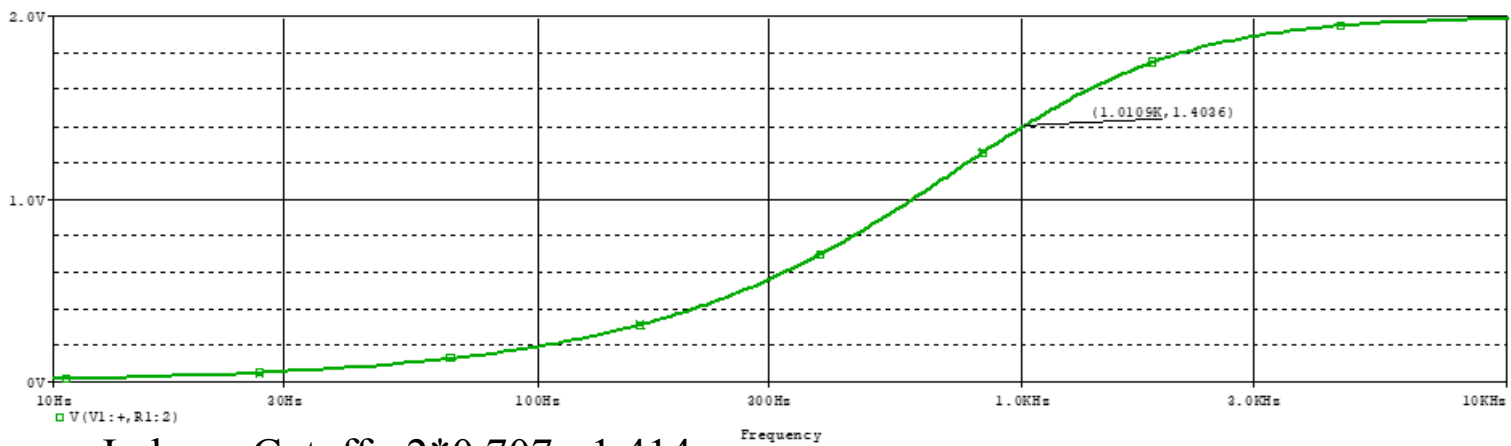
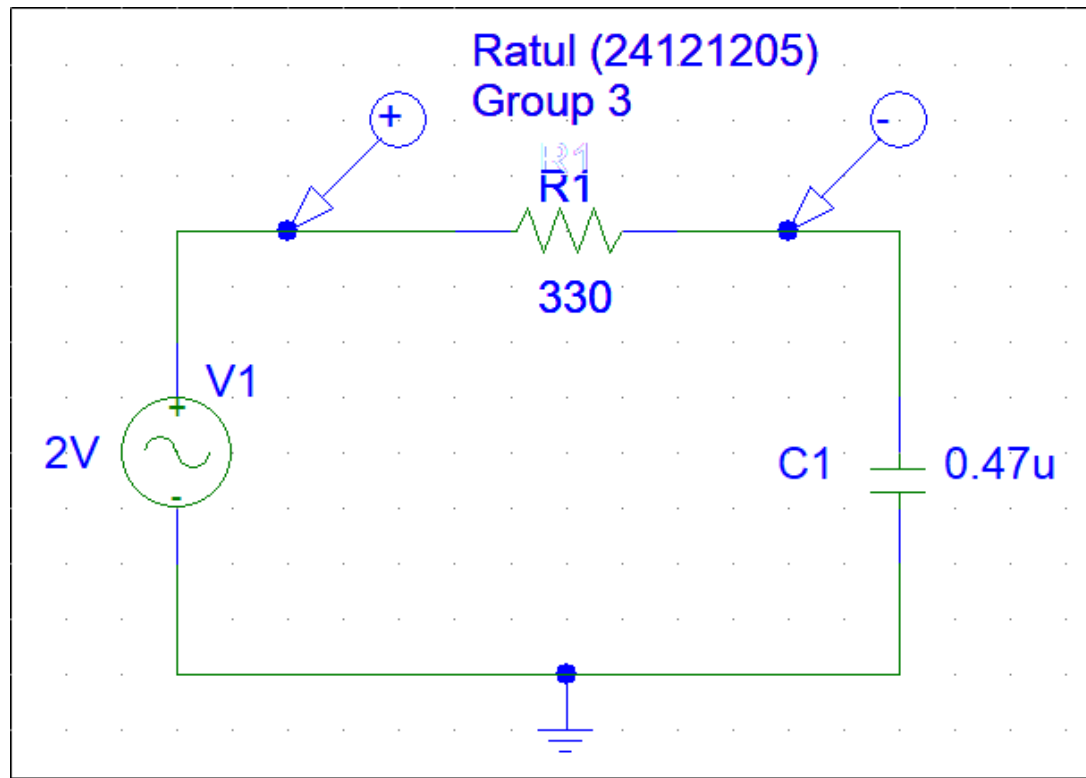
Cutoff from gain graph,



Cutoff from phase

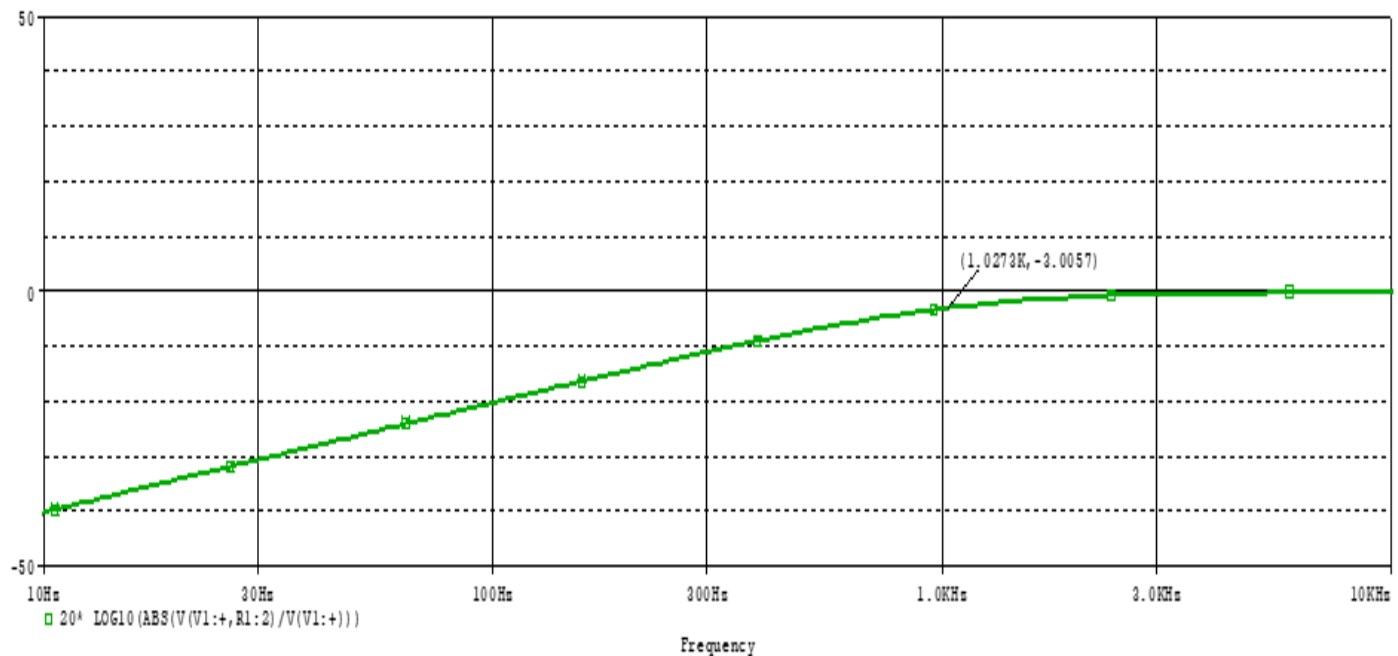
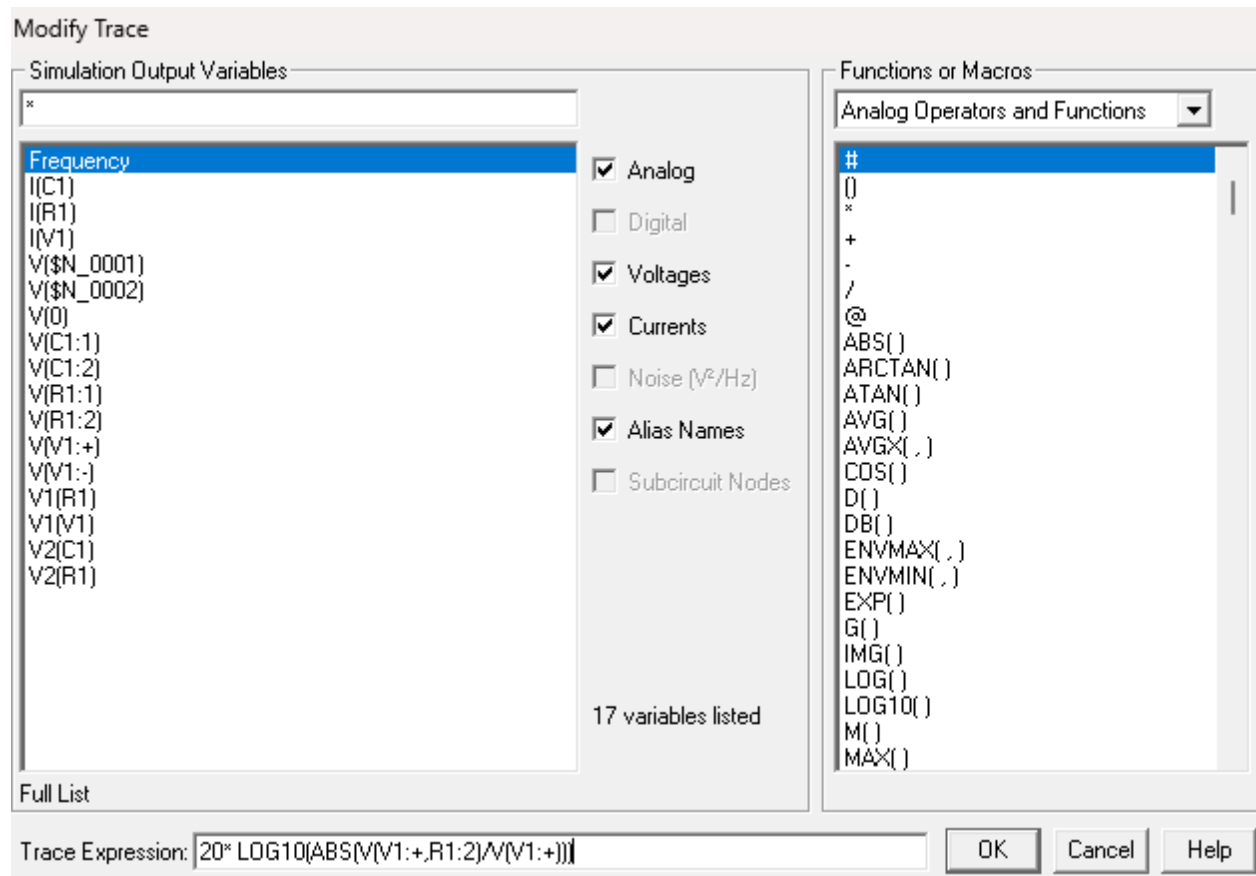


High pass Filter

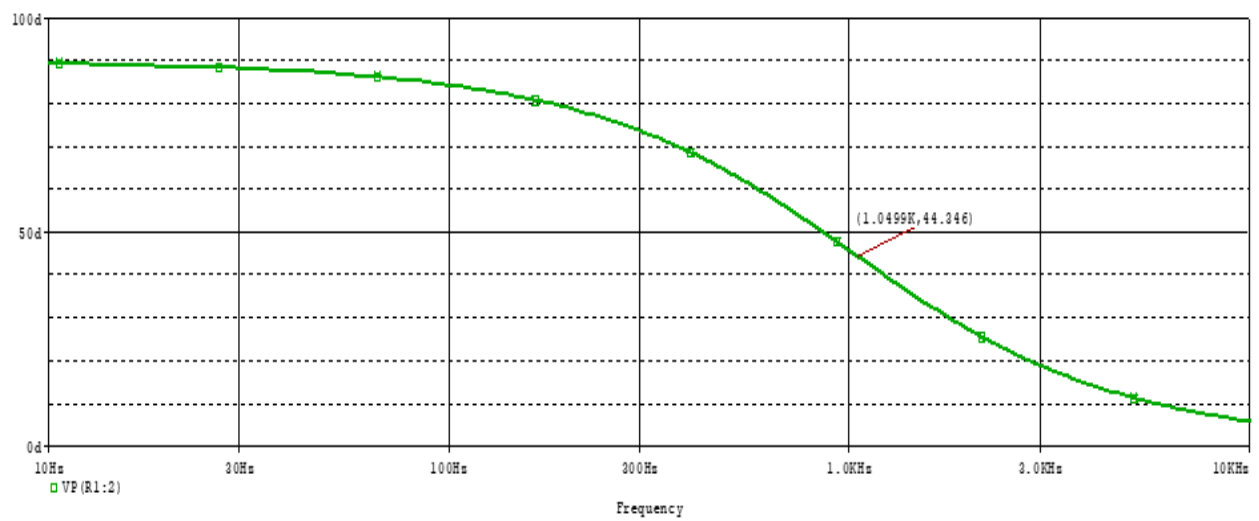
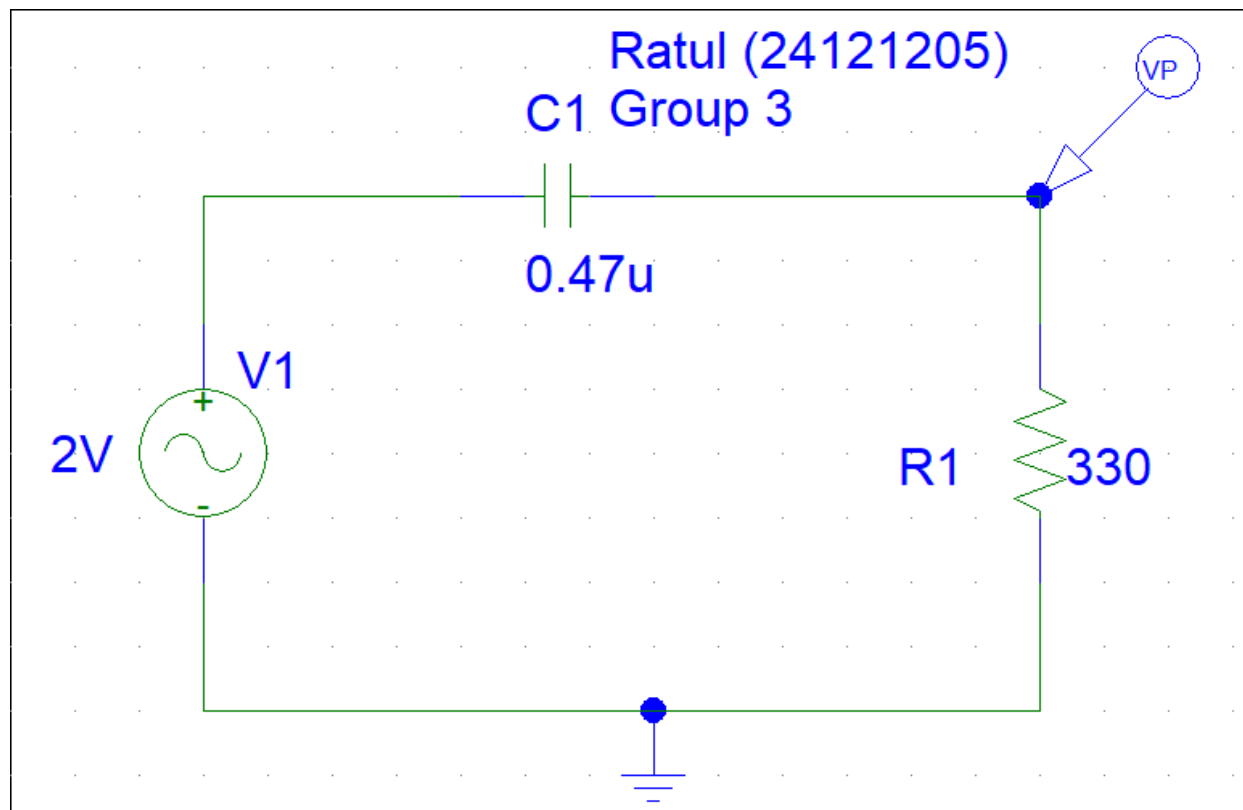


In here, Cutoff = $2 \times 0.707 = 1.414$

Cutoff from gain graph,



Cutoff from phase



Discussion of Results

In this experiment, we tested low-pass and high-pass RC filters by applying different input frequencies. The output voltages matched the theoretical response. For the low-pass filter, low frequencies passed with little attenuation, while higher frequencies were blocked. For the high-pass filter, higher frequencies passed but lower ones were attenuated. The experimental cutoff frequencies were close to the calculated values. The gain vs frequency plots clearly showed the selective behavior of the filters.

Conclusion

This experiment proved that RC passive filters can be used to control signal frequencies. The low-pass filter allowed only low frequencies while blocking high ones, and the high-pass filter did the opposite. The measured cutoff frequencies were very close to the theoretical values, which confirms the accuracy of the formula $f_c = 1/2\pi RC$

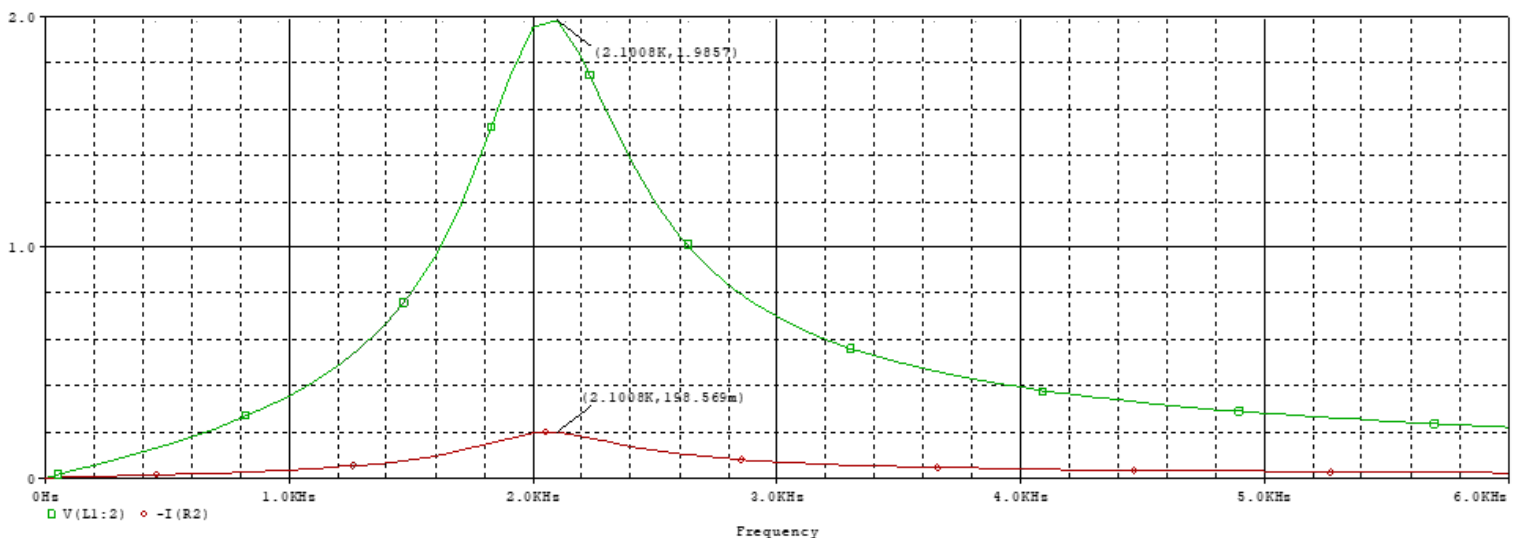
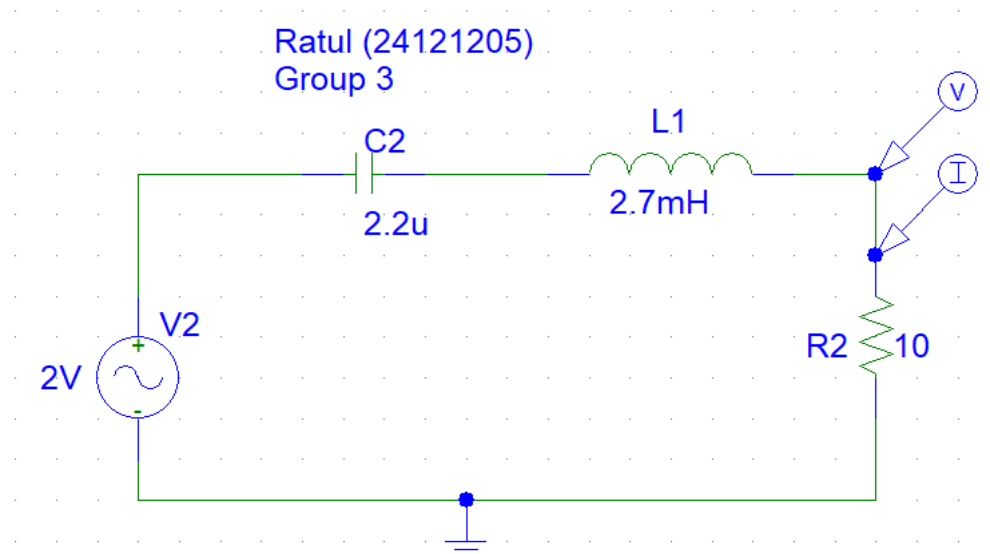
Experiment No. 6

Familiarization with Series Resonance in AC Circuits

Objective:

The purpose of this experiment is to understand how resonance occurs in a series resonance in a series RNC circuit and what its effects are on current, voltage and power.

Circuit Diagram:



Discussion of Results

In the series RLC circuit, resonance was observed when the inductive and capacitive reactances became equal. At this point, the current reached its maximum value, and the impedance was minimum (almost equal to the resistance). The voltage across the inductor and capacitor became much higher than the source voltage due to the resonance effect.

Conclusion

The experiment clearly showed the effect of resonance in a series RLC circuit. At resonance, the current reached its maximum because the impedance became minimum, equal to only the resistance. The voltages across the inductor and capacitor were much higher than the source voltage, proving the energy exchange between L and C.

Experiment No. 7

Cut-off Frequencies, Bandwidth & Selectivity

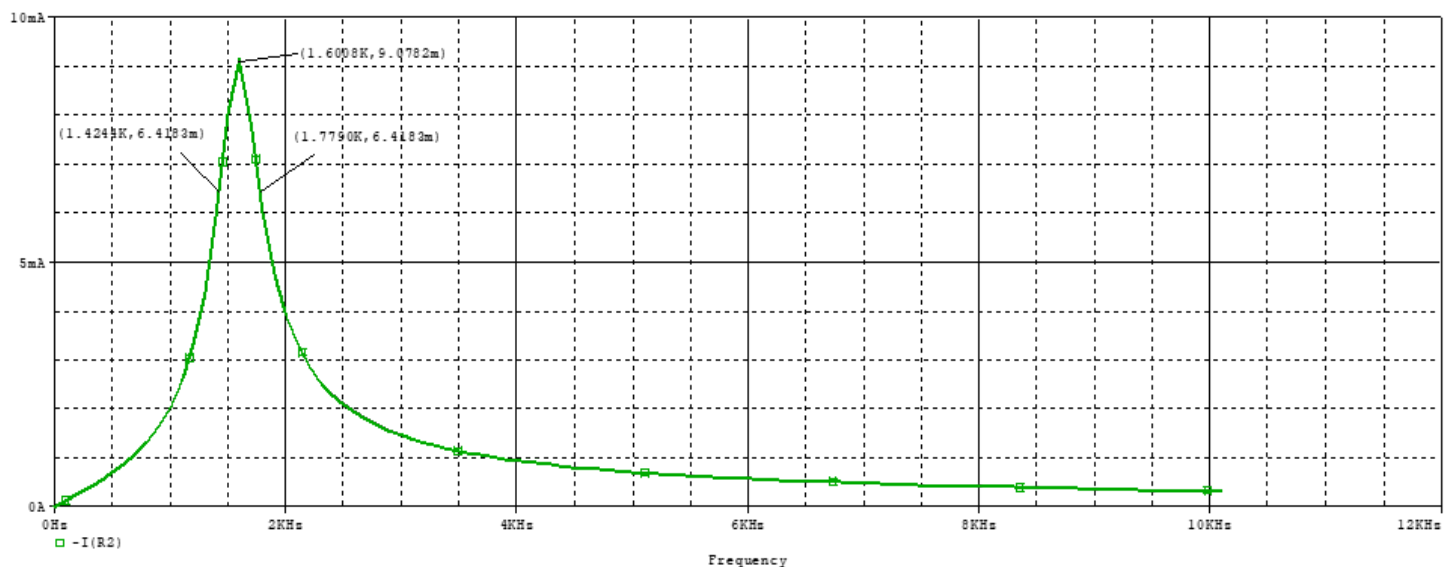
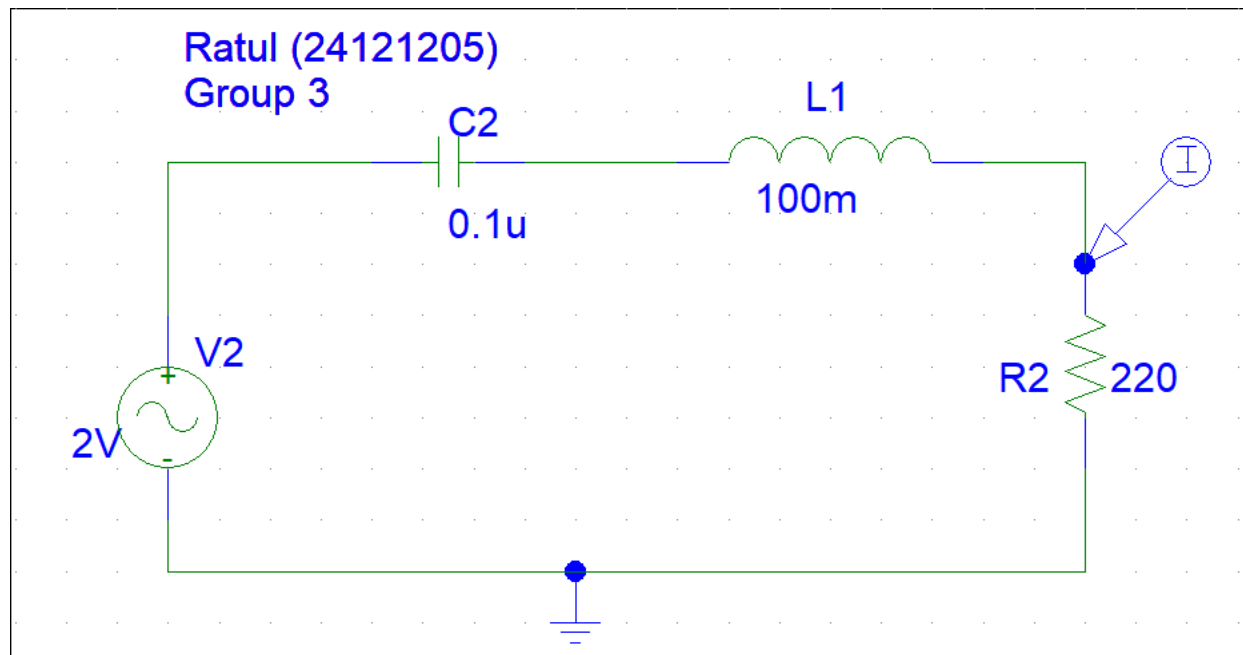
Objective:

The goal experiment is to measure the bandwidth and Q-factor of a series RLC circuit at resonance.

Circuit Diagram:

$R=220\ \Omega$

For Current



Data for Current

$I_{\max} = 9.0782 \text{ mA}$

$0.707 I_{\max} = 6.4182874 \text{ mA}$

$f_{cl} = 1.4244 \text{ kHz}$

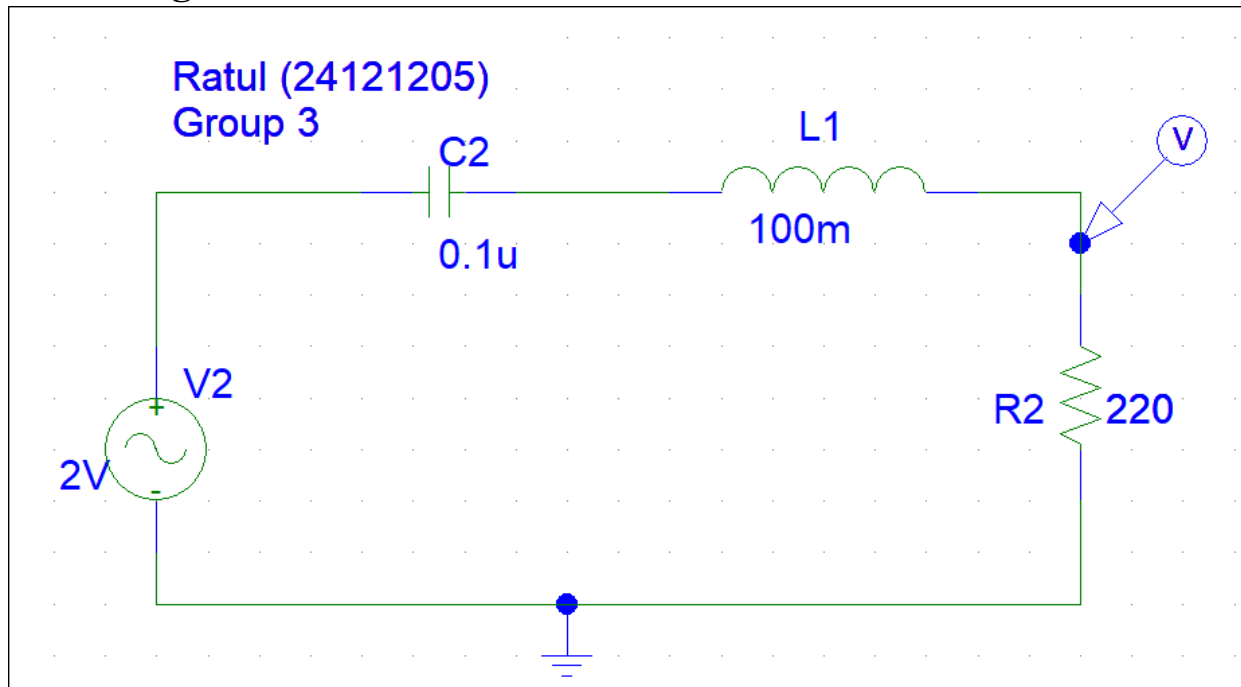
$f_{ch} = 1.7790 \text{ kHz}$

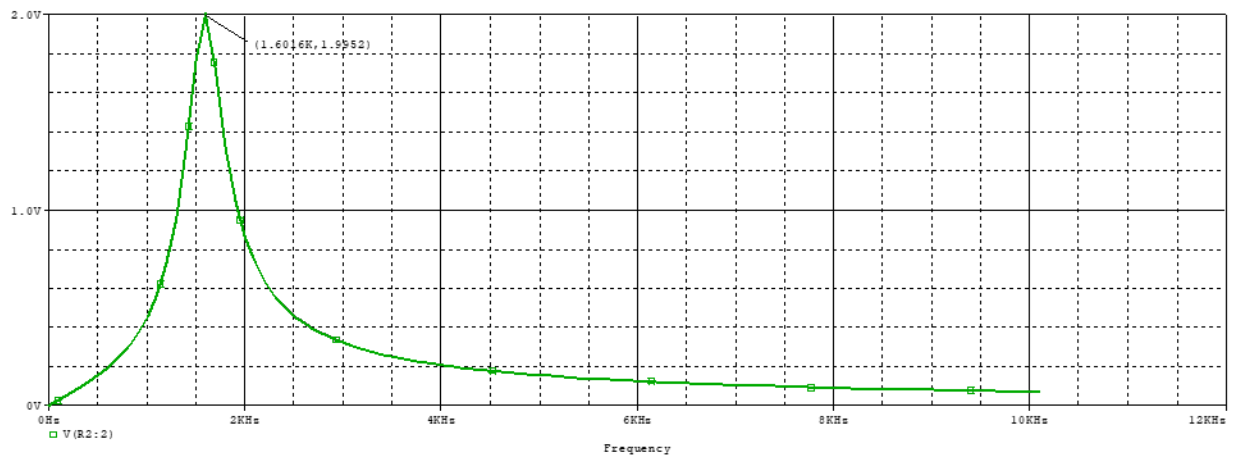
$BW = 0.3546 \text{ kHz}$

$f_r = 1.6008 \text{ kHz}$

$Q = 4.514$

For voltage

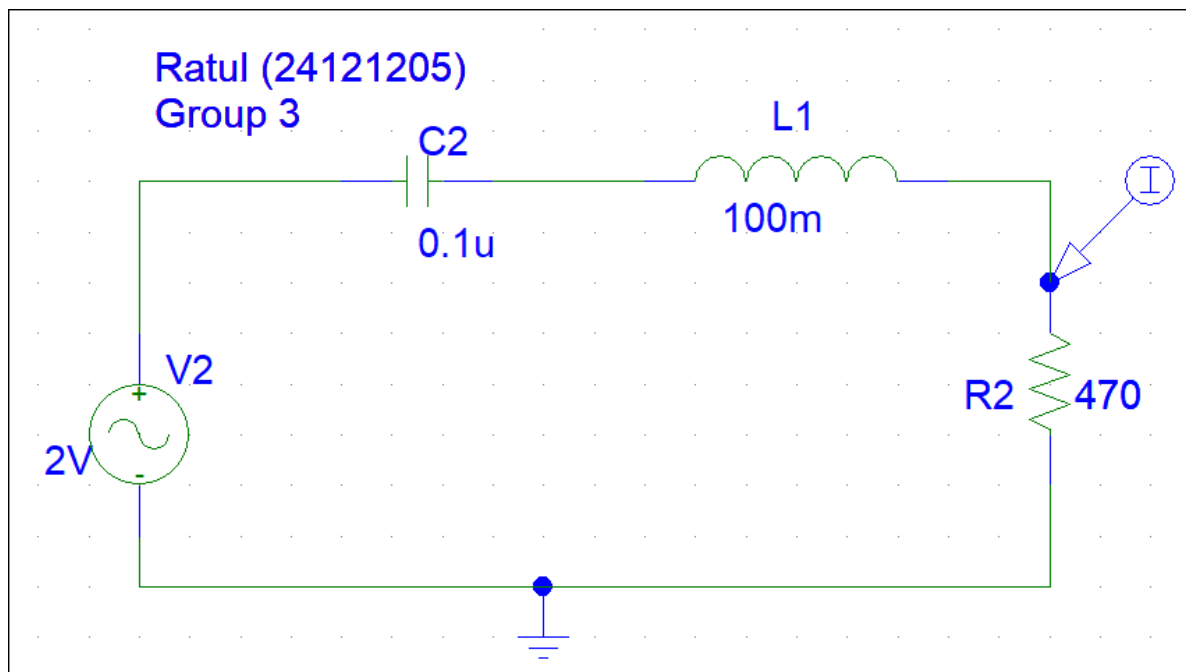


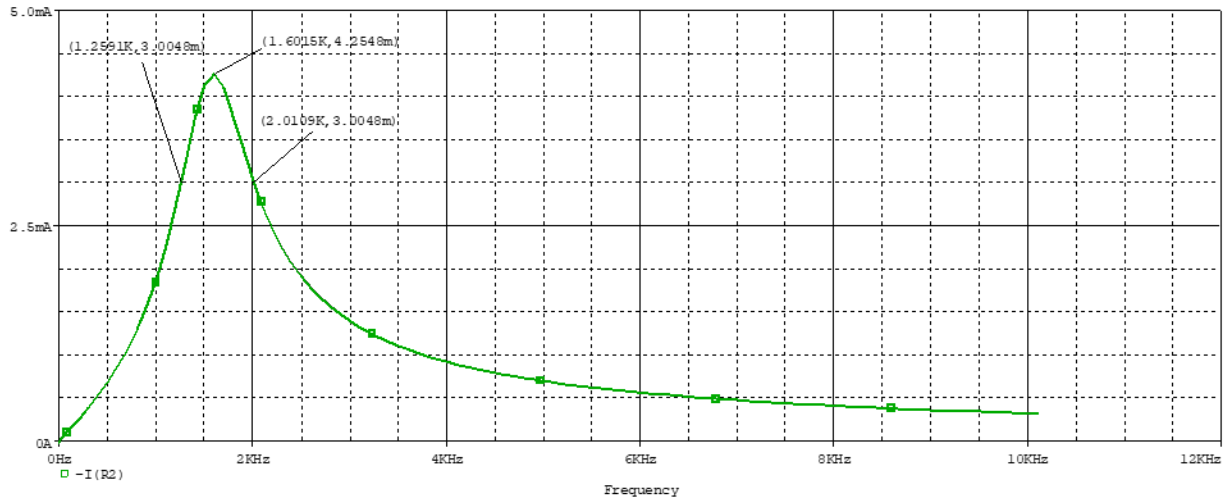


$f_r = 1.6016 \text{ kHz}$

$R = 220 \, \Omega$

For Current





Data for Current

$I_{max} = 4.2548\text{mA}$

$0.707I_{max} = 3.0081436\text{mA}$

$f_{cl} = 1.2591\text{kHz}$

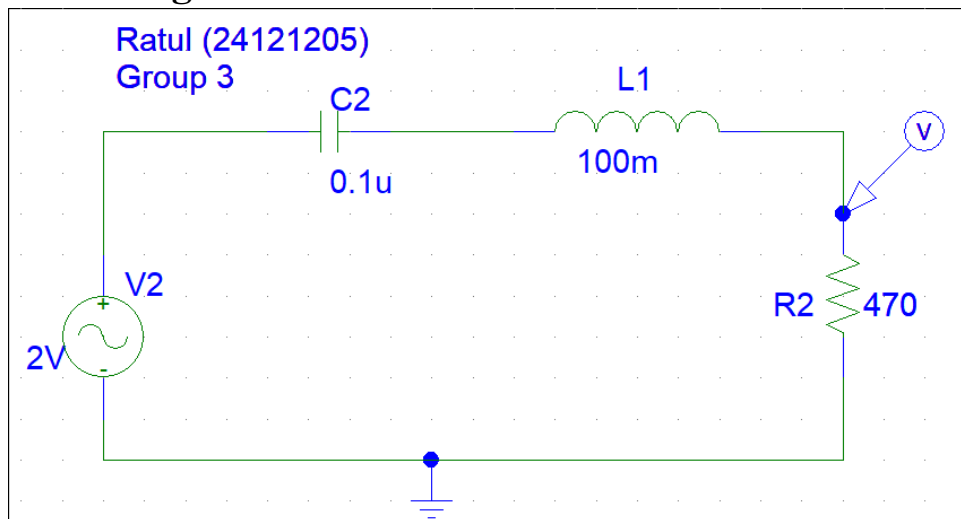
$f_{ch} = 2.0109\text{kHz}$

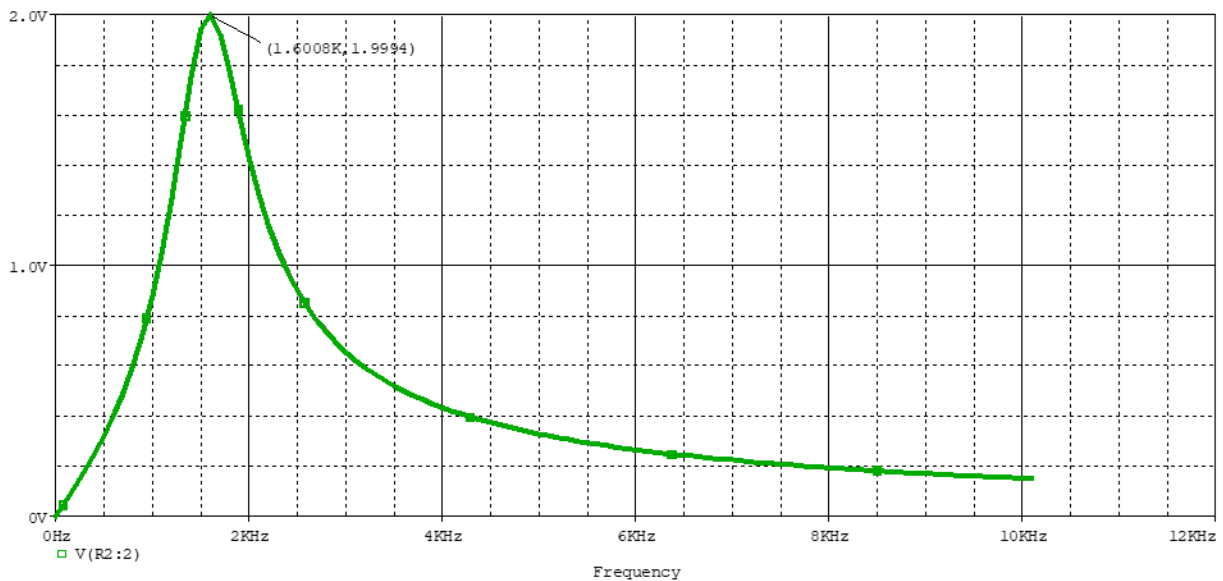
$BW = 0.7518\text{ kHz}$

$f_r = 1.6015\text{kHz}$

$Q = 2.1302$

For voltage





$$f_r = 1.6008 \text{ kHz}$$

Discussion of Results

From the experiment, the resonance frequency was found close to 1.6 kHz for both current and voltage measurements. Using the values of lower and upper cutoff frequencies, the bandwidth (BW) and quality factor (Q) were calculated. For current measurement, $Q \approx 4.51$, and for voltage measurement, $Q \approx 2.13$. These results show that the circuit's selectivity depends on the Q-factor, where higher Q gives a sharper resonance.

Conclusion

This experiment demonstrated how cutoff frequencies can be used to find the bandwidth and quality factor of a resonant circuit. The resonance frequency was close to 1.6 kHz, and the calculated values of Q showed how selective the circuit is. A higher Q means the circuit can pick a narrow range of frequencies, while a lower Q gives a wider range. This practical observation proved the theoretical relationship between resonance, bandwidth, and selectivity, which is important in filters, oscillators, and communication systems.

